THE THEOREM ON FORMAL FUNCTIONS OVER A COMPLETE NOETHERIAN BASE, WITH THE COMPLETION ON THE LEFT HAND SIDE REDUNDANT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The theorem on formal functions. Same chapter spaces-cohomology.tex, Theorem \ref{theorem-formal-functions} (cited at proof line 4494). It is the general statement whose completion of the cohomology modules is being removed in the case of a complete base ring.
- Finiteness of the cohomology of a coherent sheaf on a proper space over a Noetherian affine base. Same chapter spaces-cohomology.tex, label lemma-proper-over-affine-cohomology-finite (invoked at proof line 4497). It makes $H^p(X, \mathcal{F})$ a finite A-module, hence complete for the I-adic topology.
- A finite module over a Noetherian complete ring is complete. Algebra, label algebra-lemma-completion-tensor (invoked at proof line 4498). It shows that the completion of the left hand side is not necessary, which yields the displayed identification.

Source reference. Original lemma 00MA: sources/stacks-chow/tags.txt, source lines 816–816. Navigation pointer for \ref{algebra-lemma-completion-tensor} used inside the counted span (line 4498); the argument that consumes it is retained in the span.

Source reference. Original lemma 08AS: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 4012–4012. Navigation pointer for \ref{lemma-proper-over-affine-cohomology-finite} used inside the counted span (line 4497); the argument that consumes it is retained in the span.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original theorem 08AZ: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 4425–4425. Navigation pointer for \ref{theorem-formal-functions} used inside the counted span (line 4494); the argument that consumes it is retained in the span.

Lemma 0.1. *Let A be a ring. Let $I \subset A$ be an ideal. Assume A is Noetherian and complete with respect to I. Let $f : X \to \mathrm{Spec}(A)$ be a proper morphism of algebraic spaces. Let $\mathcal{F}$ be a coherent sheaf on X. Then*

$$H^p(X, \mathcal{F}) = \lim_n H^p(X, \mathcal{F}/I^n\mathcal{F})$$

for all $p \geq 0$.

Proof. This is a reformulation of the theorem on formal functions (Theorem 08AZ) in the case of a complete Noetherian base ring. Namely, in this case the A-module $H^p(X, \mathcal{F})$ is finite (Lemma 08AS) hence I-adically complete (Algebra, Lemma 00MA) and we see that completion on the left hand side is not necessary. □